

A comprehensive study to the protection of digital library readers' privacy under an untrusted network environment

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Abstract

Purpose – First, the authors analyze the key problems faced by the protection of digital library readers' data privacy and behavior privacy. Second, the authors introduce the characteristics of all kinds of existing approaches to privacy protection and their application limitations in the protection of readers' data privacy and behavior privacy. Lastly, the authors compare the advantages and disadvantages of each kind of existing approaches in terms of security, efficiency, accuracy and practicality and analyze the challenges faced by the protection of digital library reader privacy.

Design/methodology/approach – In this paper, the authors review a number of research achievements relevant to privacy protection and analyze and evaluate the application limitations of them in the reader privacy protection of a digital library, consequently, establishing the constraints that an ideal approach to library reader privacy protection should meet, so as to provide references for the follow-up research of the problem.

Findings – As a result, the authors conclude that an ideal approach to reader privacy protection should be able to comprehensively improve the security of all kinds of readers' privacy information on the untrusted server-side as a whole, under the premise of not changing the architecture, efficiency, accuracy and practicality of a digital library system.

Originality/value – Along with the rapid development of new network technologies, such as cloud computing, the server-side of a digital library is becoming more and more untrustworthy, thereby, posing a serious threat to the privacy of library readers. In fact, the problem of reader privacy has become one of the important obstacles to the further development and application of digital libraries.

Keywords Digital library, Information service, Reader privacy, Data privacy, Behavior privacy

Paper type Literature review

1. Introduction

A digital library is a special library constructed based on computer network (Internet), where all kinds of books and documents are of digitalization, and all kinds of information services and management are of virtualization (Han *et al.*, 2019; Wu *et al.*, 2021a), i.e. a digital library is a virtual library without wall, a shareable and expandable knowledge system constructed based on computer network and a distributed and easy to use digital information platform with no limits of time and space (Pyrounakis *et al.*, 2017). With the rapid development of network technology, the scope of a digital library is being continuously extended, which has spread all aspects of human society and become an important part of people's daily life in the

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digital information age (Mohammad, 2017). In fact, a digital library has become an essential information guarantee in today's knowledge society, whose construction level has become an important index to the national information infrastructure (Beasley, 2017).

However, in the process of supplying knowledge information services for readers, a digital library would inevitably in private collect an amount of sensitive privacy data such as readers' identities, personal interests, social relations and behavior preferences (Klinefelter, 2016; Kiran *et al.*, 2012). Moreover, with the widespread applications of emerging network technologies, such as cloud computing, in digital libraries (Shen *et al.*, 2017), the server-sides of digital libraries are becoming more and more untrustworthy, consequently, leading to a serious threat to the security of readers' privacy data (Wu, 2018; Ge, 2019). It is generally known that the right to reader privacy is one of the core values of a library, so it has to be well guaranteed (Luo, 2018). Therefore, the problem of reader privacy has become one of the most important obstacles to further development and application of digital libraries in an emerging network environment, such as cloud computing (Ma *et al.*, 2016; Wan *et al.*, 2016).

1.1 Policy limitations

The problem of reader privacy has become one of the most prominent contradictions in current library management (Han *et al.*, 2016). Moreover, with the strengthening of people's awareness of rights maintenance and the extension of scope of digital library, this problem is becoming more and more prominent (Devika *et al.*, 2012; Alistair *et al.*, 2012). Laws and regulations, due to the strong guidance, have always been a kind of powerful means to protect civil rights (Augus, 2009). To this end, since the middle and end of last century, in the library community, it has been actively engaged in the formulation of policies relevant to the protection of reader privacy (Klinefelter, 2016; Lin, 2019; Liang *et al.*, 2014; Xiao, 2014) and has been worked hard to promote legislatures to issue laws and regulations on the protection of library reader privacy.

The USA is the first country to pay attention to the problem of library reader privacy. In the US Constitution, the protection of reader privacy is clearly stipulated (Beck, 2002). It views reader privacy as an important part of human rights and issues special laws and regulations on privacy protection. For example, in 1978, under the promotion of the American Library Association (i.e. ALA), the first state legislation Library Record Confidentiality Law was issued officially, where readers' privacy rights that a library should protect have been clearly defined (Xiang *et al.*, 2018). Later, the American online alliance (i.e. AOL) issued the Online Privacy Guide in 1998, which is used to protect readers' identifiable privacy information collected by a website. It stipulates that each website should not only fully inform readers of its data collection behaviors but also inform them of the specific uses of data collection (Flores *et al.*, 2014). Also, the ALA formulated the ALA Privacy Policy in 2003 (revised in 2008), which promises that the user information should not be collected by a website, unless it is chosen to provide by users, and even for the information voluntarily provided to the website by users, it can only be used to improve the information service by the website (Flores *et al.*, 2014). In addition, for the problem of general privacy protection, the European Union issued the General Data Protection Regulation (GDPR) in 2018, which stipulates a set of data management rules that enterprises have to abide by, which can well enhance the privacy of user data. In summary, in developed countries (especially, the USA), the policies on reader privacy are relatively comprehensive, which are helpful to solve the problem of digital reader privacy under an untrustworthy cloud environment.

In China, the legislation on the protection of library reader privacy has also been paid attention. The Public Library Law of the People's Republic of China, which was began to formulate in 2008 and then officially issued in 2018, is the first law in China specifically for strengthening the management of public libraries, promoting the further development of

public libraries and better protecting people's public reading rights (Li, 2017). In Article 43, it is clearly stated that personal information, lending information and other information that may involve readers' privacy should be properly protected by a public library, which should not be sold or provided to others illegally. However, compared with developed countries, in Chinese library community, it is still in its infancy in the protection of reader privacy, the relevant legal provisions are not detailed and comprehensive enough and the protection of people's online privacy is still at a lower level (Wang, 2018), which still needs to be further refined and improved at the level of legislation and trade policies.

Although laws and regulations can alleviate the problem of digital library readers' privacy under a cloud environment, they cannot solve the problem fundamentally (Klinefelter, 2016; Han *et al.*, 2016; Wu *et al.*, 2018a). It is because for laws and regulations, their effectiveness is based on the precondition that they can be strictly observed by the library and its administrators, i.e. based on the assumption that the server-side of a digital library is highly trusted. However, a large number of privacy leakage incidents (e.g. the incident of Google employees snooping minors' email records and the incident of Swiss bank leaking customers' information) show that driven by interests, the possibility of system administrators snooping readers' privacy information is widespread (Wu *et al.*, 2012, 2018b). Especially, with the wide application and popularization of emerging network technologies, such as cloud computing, the servers of digital libraries are widely deployed at the untrustworthy cloud (Tian *et al.*, 2017), which further weakens the control and management of a digital library to readers' privacy information, reduces the credibility of the server-side of a digital library and as a result, increases readers' privacy leakage risk. In summary, we can conclude that the protection of digital library readers' privacy not only needs to be improved by laws and regulations but also needs to be solved by advanced technical methods.

1.2 Technical challenges

The privacy interpretation of Library Bill of Rights issued by the ALA defines readers' right of privacy as the right of readers to have their personal data and behavior preferences to not be monitored by others. According to this, we can divide the reader privacy of digital library into two main categories (Xiang, 2014; Wu *et al.*, 2018c).

- (1) Data privacy, including reader identity data (such as ID number, phone number and home address) and reader characteristic data (such as occupation, income, gender and age). This kind of privacy data are usually published by readers themselves after registration.
- (2) Behavior privacy, referring to the personal interest preferences within reader behaviors (such as the reader favorite literature categories behind information retrieval behaviors), when readers use the information services (such as literature retrieval, download, lending and reading services) provided by a digital library.

In general, readers' behavior privacy information is collected in private by the server-side of a digital library (to provide better personalized information services for readers and in turn improve the service quality), which may be not known by readers. It is obvious that a large collection of the reader sensitive privacy information by the untrustworthy cloud server-side of a digital library would pose a serious threat to reader privacy (Yi *et al.*, 2014).

The protection for reader data privacy is similar to the traditional problem of data confidentiality. Since traditional security technical strategies (such as identity authentication (Ding *et al.*, 2018) and access control (Yang *et al.*, 2018)) can effectively prevent unauthorized data access from external users of a system, it seems that they can be used to solve the problem of digital library readers' data privacy under an untrustworthy cloud environment. However, for the traditional technical strategies, their effectiveness is built based on the assumption that

the server-side of a digital library is highly trusted, i.e. they are only targeted for the external illegal users of a digital library system and cannot prevent the internal managers of the untrustworthy server-side (or the attackers who has controlled the server-side) from accessing reader privacy data stored in the library database (Lu *et al.*, 2013). In order to ensure the confidentiality of reader privacy data stored in the untrustworthy server-side, one of the most immediate ways is data encryption (by encrypting reader privacy data in a trusted library client and then storing it to the server-side). However, for this kind of methods, how to realize secure and efficient query operations over the encrypted data is a huge challenge (i.e. the problem of querying encrypted data). In summary, under an emerging network environment, the server-side of a digital library system is untrustworthy (Wu *et al.*, 2018c), and the security problem of readers' privacy information on the server-side cannot be solved immediately by traditional data security technical means (such as identity authentication, access control and data encryption).

The second kind of readers' behavior privacy refers to readers' interest preference known by the server-side of a digital library by collecting and analyzing various behaviors (i.e. information service requests) from the readers (Xiang, 2014; Wu *et al.*, 2018c). Generally, the protection of reader behavior privacy is more challenging than that of reader data privacy. First, traditional security technical methods, such as data encryption, (Mei *et al.*, 2018) are not suitable for the protection of reader behavior privacy. This is because each kind of information services supplied by a digital library is closely related to the support of the server-side. As a result, if the data of readers' service requests are encrypted by an encryption algorithm, then the server-side would not be able to correctly understand the service requests, resulting in the unavailability of information services. Second, the information services provided by a digital library system for readers are of various forms (such as information retrieval, file download, literature reading and lending), consequently, leading to various reader privacy leakage channels existing in a digital library system. Moreover, there are generally strong semantic correlations among all kinds of information service requests issued by the same reader during a certain period of time (Wu *et al.*, 2018c), making it unsuitable to protect each kind of information service requests independently. In addition, it is also required that the protection of reader behavior privacy should not compromise the usability of existing information services (including no changes to the efficiency and accuracy of information services and no changes to the existing platform architecture and service algorithm). All of the above reasons result in the high complexity of solving the problem of reader privacy protection in a digital library system, i.e. it is difficult to be directly solved by a traditional technical method.

1.3 Contributions

In the field of library sciences, although there have been some research attempts (Wu *et al.*, 2018a, c, 2019a, b, 2020a; Shao *et al.*, 2014) to solve the problem of reader privacy protection in a digital library from the perspective of technology, these technical methods are more trying to use existing security means (such as authentication, access control and data encryption) to improve the security of readers' privacy information, and they are not comprehensive and systematic enough. As we know, this paper is the first survey on the protection of reader privacy of a digital library under an untrustworthy network environment, which presents a valuable attempt to analyze and discuss the problem of reader privacy protection of a digital library system from a technical point of view. Specifically, the main contributions of this paper are threefold.

- (1) This paper deeply analyzes the problems and challenges faced by the protection of digital library reader privacy under an untrustworthy network environment. Then, it

comprehensively formulates the requirements that the protection of data privacy and the protection of behavior privacy should meet in a digital library system.

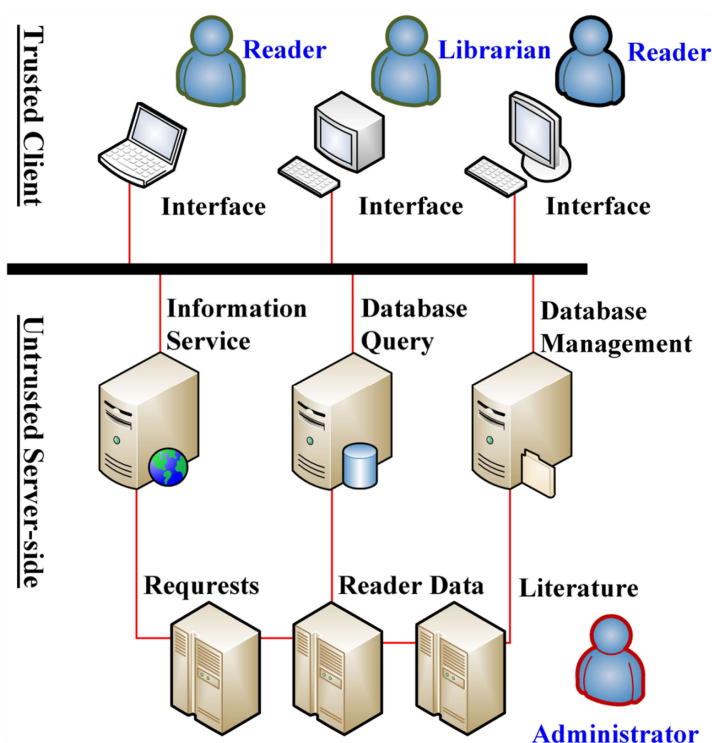
- (2) This paper comprehensively reviews the technical characteristics for each kind of existing information security methods and then deeply analyzes their application limitations in the protection of readers' data privacy and behavior privacy under an untrustworthy network environment.
- (3) According to the requirements of the protection of reader data privacy and behavior privacy, this paper comprehensively compares the advantages and disadvantages of all kinds of existing data security methods from the perspectives of usability and security and then presents several key problems that should be taken into consideration in the protection of digital library readers' privacy, as a result, providing some references and directions for the follow-up research.

The rest of this paper is organized as follows. [Section 2](#) presents the overall framework of a digital library system and then describes the protection needs of readers' data privacy and behavior privacy. [Section 3](#) reviews the technical features of such kinds of data security methods as identity authentication, access control and encryption and then analyzes the limitations of applying them for the protection of reader data privacy. [Section 4](#) reviews the technical characteristics of such kinds of privacy methods as anonymity, obfuscation, confusion and encryption and then analyzes the limitations of applying them for the protection of reader behavior privacy. [Section 5](#) compares the advantages and disadvantages of each kind of existing privacy methods from the perspectives of usability and security and then formulates several key problems that should be taken into account in the protection of digital library readers' privacy, consequently, providing some references and directions for the follow-up research. [Section 6](#) summarizes this work.

2. Problem statement

In [Figure 1](#), we present the general framework of a digital library system. It can be seen that the system is based on a client/server architecture, where each reader can submit his personal data or information service requests through a client interface of digital library to the server-side, so as to realize data storage or obtain information services (such as retrieval, download, reading and lending). It is noted that in a digital library system, the server-side is untrustworthy, while each client is trusted because it is completely controlled by readers themselves. In the system, there are three kinds of participants, i.e. readers, librarians and administrators.

- (1) Readers (the owners of privacy data), at registration, who submit a large number of personal sensitive data (including personal identity data and characteristics data, called reader profile data) to the server-side of a digital library. Moreover, readers issue various service requests to the server-side to obtain corresponding information services, and the request data (called reader behavior data) would be collected by the untrustworthy server-side automatically, as a result, posing a threat to the privacy of reader behavior.
- (2) Librarians (the workers of the library), who completes their general management work (e.g. the management of documents) through a client management interface of a digital library. We assume that librarians are credible, since they are professional library staff with good occupation ethics, and more importantly, a digital library system can control the management permissions of each kind of librarians through



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Figure 1.
The basic framework
of a digital library
system

traditional information security strategies, such as identity authentication and access control, so that each librarian can only access limited reader data.

- (3) Administrators (the manager of the server-side of a digital library), who are responsible for the installation, configuration and management of the whole digital library system.

It can be seen that under an untrustworthy network environment, such as cloud computing, traditional information security technical means used in a digital library cannot restrict the administrators of the library server-side, i.e. they can access the reader privacy data stored in the server-side without restriction. In addition, there are also attackers independent of the system, which mainly target the server-side of the system, i.e. once they conquer the server-side, they can obtain the same privileges as the administrators. Therefore, the administrators (attackers) in the server-side are considered to be untrustworthy, and they are a key cause of reader privacy leakage (i.e. the biggest potential attackers).

The problem of digital library reader privacy concerned in this paper mainly refers to the confidentiality of reader privacy information, so in the privacy model used by this paper, it is assumed that the attackers are passive (honest but curious), whose main interest is to spy and steal the reader privacy information on the server-side of a digital library, while other active malicious behaviors (such as tampering and destroying data, replay attack and denial of service attack) are beyond the scope of this paper. In addition, bypass attack and physical attack are also out the scope of this paper (since they can be well solved by existing security techniques).

Now, combining with [Figure 1](#), we can see that the biggest challenge to overcome the problem of reader privacy is how to improve the security of reader privacy data (profile data and behavior data) on the untrustworthy server-side of a digital library, under the precondition of not compromising the usability of existing information services. In the following two subsections ([Sections 2.1 and 2.2](#)), based on the system framework shown in [Figure 1](#), we will describe the protection needs for readers' data privacy and behavior privacy, respectively.

2.1 Data privacy needs

In order to ensure the confidentiality of privacy data stored in the untrustworthy server-side of a digital library, data encryption is one of the most immediate ways, i.e. to encrypt reader privacy data in a trusted client and then submit it to the untrustworthy server-side for storage, so that even if the encrypted data are leaked, it is difficult to be understood by an attacker, thereby, ensuring the privacy and security of reader data. However, in a digital library system, there are a great number of query operations defined on reader privacy data. Therefore, how to realize secure and efficient query operations on the encrypted data is a huge challenge. Combined with the general framework and data processing process shown in [Figure 1](#), we formally describe the requirements that an ideal approach to the protection of reader data privacy should meet.

Constraint 1 (Accuracy): After the introduction of the privacy protection approach, all kinds of query operations (such as similar query, range query and join query) defined on reader privacy data submitted by users through a digital library client still can be correctly executed on the server-side database, and the query result returned from the server-side to the client have to be a superset of the real query result, so as to ensure the accuracy of user query operations.

Constraint 2 (Efficiency): Before and after the introduction of the privacy protection approach, the execution time consumed by a reader data query operation from a digital library client should be not decreased (or not decreased significantly), so as to ensure the efficiency of user query operations.

Constraint 3 (Practicality): The practicality of readers' data privacy in a digital library system can be divided into the following three levels.

- (1) Level I practicality. It should require no change to the architecture of the whole digital library system.
- (2) Level II practicality. It should require no change to readers' using patterns.
- (3) Level III practicality. It should require no change to the existing information management system of a digital library system.

Constraint 4 (Security): Because the server-side of a digital library is untrustworthy, an effective approach to the protection of reader data privacy should ensure the confidentiality of reader privacy data stored in the server-side, i.e. it should be difficult for attackers of the server-side to obtain reader profile data; and even if the reader data have been obtained, it should be still difficult for attackers to further know the privacy information behind the reader data.

2.2 Behavior privacy needs

Because each kind of information services (such as information retrieval and recommendation) exists as an important part of a large-scale digital library platform, it is

not practicable to change the whole platform framework due to reader privacy protection. Therefore, an effective approach to the protection of reader behavior privacy should be completely transparent to the service algorithms on the server-side and each client interface of a digital library (i.e. no changes to the service algorithms and users' usage patterns). In addition, readers' behavior privacy protection also should not compromise the accuracy and efficiency of information services. Therefore, an ideal privacy protection approach to reader behaviors should satisfy the actual needs of each kind of digital library information service in terms of practicality, accuracy, efficiency and security. Below, combined with the basic framework and data processing process shown in [Figure 1](#), we formally describe the requirements that an ideal approach to the protection of reader behavior privacy should meet.

Constraint 5 (Accuracy): After the introduction of the privacy protection approach, the information service result returned from the server-side of a digital library has to be equal to the real service result or a superset of the real service result. This constraint ensures that each reader can finally obtain an accurate information service result.

Constraint 6 (Efficiency): After the introduction of the privacy protection approach, the scale of information service result returned from the server-side has to be linearly related to that of the real service result (i.e. the result corresponding to the original service request). This constraint ensures that the reader behavior privacy protection would not lead to a significant impact on the execution efficiency of an information service.

Constraint 7 (Practicality): The practicality of protection of readers' behavior privacy mainly includes two aspects, i.e. no changes to the usage pattern of each client user and no changes to the information service algorithm at the server-side of a digital library.

According to the theory of human behavior, we know that in a digital library, a sequence of service requests issued by the same reader during a certain period of time usually has a regular characteristic distribution (e.g. all around some fixed topics). Moreover, there is strong semantic relevance between all kinds of behavior privacy behind the service requests from the same reader. For example, for the book retrieval behaviors and book reading behaviors from the same reader, they are likely to be around similar topics. The regular characteristic distributions require that behavior privacy protection should not in isolation take into account the behavior privacy contained in one service request or one kind of service requests. To this end, we describe three levels of security of digital library readers' behavior privacy.

Constraint 8 (Security): The security of readers' behavior privacy in a digital library system can be divided into the following three levels.

- (1) Level I security. According to service request data with single type submitted by a reader in a single information service activity, it is difficult for the untrustworthy server-side to accurately infer the reader interest preferences (i.e. reader behavior privacy).
- (2) Level II security. Under the premise of meeting the Level I security, by analyzing the service request sequence of single type submitted by a reader in multiple information service activities during a certain period of time, it is difficult for the untrustworthy server-side to accurately infer the reader behavior privacy.
- (3) Level III security. Under the premise of meeting the Level II security, by analyzing the service request sequences of multiple types submitted by a reader in multiple information service activities during a certain period of time, it is difficult for the untrustworthy server-side to accurately infer the reader behavior privacy.

It can be seen that the requirements of an ideal protection approach to reader behavior privacy can be summarized as follows. Under the basic premise of not changing the existing digital library platform architecture, not changing the existing information service algorithms on the server-side and not changing the accuracy and efficiency of information services, it is difficult for the untrustworthy server-side of a digital library to accurately infer the reader behavior preferences from the service request sequences submitted by readers, so as to comprehensively improve the security of reader behavior privacy on the untrustworthy server-side. It is deemed that the privacy right of each digital library reader is protected effectively, if and only if the data privacy and behavior privacy of the reader is protected effectively. However, existing research achievements on the protection of library readers' privacy right are more proposed from the perspective of policies and regulations ([Lin et al., 2019](#); [Liang et al., 2014](#); [Xiao, 2014](#); [Xiang, 2014](#); [Yi et al., 2014](#)), and the research achievements proposed from the perspective of technology are still lacking. In the following two sections ([Sections 3 and 4](#)), we will review and analyze the limitations of existing information security technical means in the protection of readers' data privacy and behavior privacy.

3. Data privacy methods

The reader data privacy protection in a digital library is similar to traditional data confidentiality. However, traditional security technical methods (including identity authentication, access control and data encryption) can effectively prevent external users of a system from accessing unauthorized data illegally, so as to ensure the data confidentiality. In this section, we introduce the technical characteristics of traditional security methods and analyze the application limitations of them in the protection of reader data privacy of digital library.

3.1 Database security methods

A database is a shareable data collection, where how to ensure data security is one of the most important problems that need to be solved ([Izzat et al., 2018](#)). In order to ensure data confidentiality, database systems adopt many security technical strategies, such as identity authentication ([Ding et al., 2018](#)), access control ([Yang et al., 2018](#)) and security audit ([Izzat et al., 2018](#)). User identity authentication, including user identification and authentication, is one of the most common database security technologies. In this kind of strategies, the system provides a certain way for users to identify themselves. Then, each time when any user requests to login, the system would first check the user identity and ensures that only the user who successfully passes the identification can obtain the right to use the system. At first, user authentication only has a form of password verification ([Jeff et al., 2004](#)). Now, fingerprint verification ([Hanene et al., 2013](#)), face verification ([Bianco, 2016](#)), pupil verification ([Villalobos et al., 2013](#)) and other biometric verification become more and more mature, and most of them have been applied to user authentication. As a result, user identity authentication can be divided into single factor authentication (such as password, smart card, dynamic password and bio-metrics) and multi-factor authentication ([Liu et al., 2021](#)) that combines multiple single factor authentications together to strengthen the security of user identity authentication.

Access control, through user permission definition and authorization, ensures that only the users with legal permission can access the authorized data stored in the database, while the unauthorized users cannot access the database. Therefore, the objects of access control are users, and the targets of protection are data. Access control is one of the basic security means of a database, which is of great significance to database security. Access control can be divided into two categories, i.e. discretionary access control (DAC) ([Augus, 2009](#); [Lin et al., 2019](#)) and mandatory access control (MAC) ([Liang et al., 2014](#); [Xiao, 2014](#)). In DAC, users have

access to their own data and can grant access of their data to other users (or revoke access). The security of DAC is relatively low; thus, it cannot provide strict protection for system resources (Wu *et al.*, 2010). In MAC, all users and data are assigned with certain security levels. All the users cannot change their own security levels, and only the users who follow the assigned rules can access the required data legally. Since users cannot directly change their access control attributes, the MAC model can provide stronger security protection to prevent abuse of DAC (Hayawardh *et al.*, 2012). Different from the above two kinds of technical means, security audit is a post-processing security protection activity (Julien *et al.*, 2013), where the audit function records all the user operations on a database in an audit log. With the help of the audit log, we can monitor all kinds of user behaviors in the database system and then can find out the responsible party of each security problem. Due to the high time cost and space cost of audit activities, security audit can be chosen whether to implement by database administrators according to the actual situation.

Usually, reader profile data are stored in the back-end database of a digital library. With the help of existing security technical strategies of a database system, we can prevent the access of external users to the reader profile data, thereby, improving the security of reader data privacy on the untrustworthy server-side. As these technical strategies are supported by each database system itself, the accuracy, efficiency and practicality mentioned in Section 2.1 are no longer a problem, i.e. the usability constraint of the protection to reader data privacy can be well satisfied. However, these technical strategies are all developed based on an assumption that the server-side of a digital library is highly trusted, i.e. they are only targeted for illegal users outside a digital library and cannot prevent database administrators of the untrustworthy server-side (or attackers who have controlled the server-side) from accessing the reader privacy data stored in the digital library database. In other words, the existing database security technical strategies cannot meet the security requirements mentioned in Constraint 4 (i.e. security).

3.2 Data encryption methods

Encryption is the most natural way to ensure data confidentiality, by encrypting reader privacy data on a client and then storing it on the server-side, which makes that even if the encrypted data are leaked, it is difficult for attackers to understand, so as to ensure the security of reader data privacy on the untrustworthy server-side (Lu *et al.*, 2013). However, it is challenging for a data encryption method how to realize secure and efficient query operations on the encrypted data, resulting in a cipher-text query problem. In this section, we first discuss the feasibility of treating the cipher-text query problem as full homomorphism encryption. Second, we introduce some representative cipher-text range query methods and cipher-text keyword query methods, respectively. Finally, we discuss the limitations of applying each kind of existing encryption methods to improve the security of reader privacy data on the untrustworthy server-side.

As early as 1978, after the birth of RSA (Rivest-Shamir-Adleman) system, the concept of homomorphism encryption was proposed to try to directly solve the calculation problem on cipher-text data (Tian *et al.*, 2017). Since the function of homomorphism encryption depends on the kind of operations it can support, it can be divided into three categories, i.e. (1) additive homomorphism encryption, which can support addition (such as the Paillier system (Shifa *et al.*, 2016)); (2) multiplicative homomorphism encryption, which can support multiplication (such as the ElGamal system (Nalwaya *et al.*, 2015)), and (3) fully homomorphism encryption, which can support addition and multiplication at the same time. In theory, fully homomorphism encryption can support any kind of complex operations on cipher-text data; thus, it can well solve the cipher-text query problem. However, although the concept of homomorphism encryption has been proposed in 1978, it was not until 2009 that the first

homomorphism encryption structure was designed by Gentry based on the ideal lattice theory (Gentry, 2009). At present, the performance of homomorphism encryption is far from being practical (Tian *et al.*, 2017). In addition to the performance problem, it needs to further study how to support each kind of query operations on the basis of homomorphism encryption (Naehrig *et al.*, 2011). Wang *et al.* (2012) discussed the feasibility of using homomorphism encryption to support common query operations, such as selection query, range query, joint query and aggregation query. However, it has been proved that for a data query of uncertain result scale, homomorphism encryption cannot guarantee the accuracy of the query result. Thus, before the performance of homomorphism encryption reaches the practical level, scholars have to study some methods of small performance cost to solve the cipher-text query problem (Tian *et al.*, 2017; David, 2015). However, the improvement of performance usually requires a compromise on the function. In the following sections, we introduce some representative cipher-text query methods that can only support some specific kinds of data query operations.

Range query is one kind of common database query operation (Ayanso *et al.*, 2014). In general, order-preserving encryption (Alexandra *et al.*, 2011) can support range query on cipher-text, whose biggest advantage is that the order relationship among the values of cipher-text is consistent to that of plain-text, so that each range query operation in a database system can be correctly executed over the cipher-text without transformation (Li *et al.*, 2015). However, there also exists some security problem in order-preserving encryption, especially for one-to-one order-preserving encryption, its cipher-text may reveal the order relationship of plain-text, which is vulnerable to statistical attacks (Yang *et al.*, 2017). In general, the cipher-text data of numerical type needs to support range query operations, and the cipher-text data of textual type needs to support matching query operations, where the most common one is to search all cipher-text records containing some given keywords (i.e. keyword cipher-text query operations (Curtmola *et al.*, 2011)). Here, we introduce three kinds of typical symmetric searchable encryption methods that can support keyword cipher-text query operations (see the survey (Wang *et al.*, 2016) for asymmetric searchable encryption). The first kind of encryption methods does not require the index construction (Song *et al.*, 2002), whose query processing needs to scan each record in database tables. This kind of method has almost no additional storage cost, but it usually has to traverse each word in text records, resulting in heavy time overhead, and it reveals the location and amount matched to search words (Tian *et al.*, 2017). The second kind of encryption methods requires the index construction on each textual record (Chang *et al.*, 2005; Hore *et al.*, 2012), so as to efficiently determine whether a record contains a given search term, without revealing unnecessary matching information. In such kind of method, the query workload is directly proportional to the size of a database table, but each record does not need to be scanned. The third kind of method requires the index construction on the whole database table (Curtmola *et al.*, 2011), so the query processing workload is only related to the size of a result set, independent of the size of the database table. Therefore, for this kind of method, the computation time overhead is better than those of the previous two kinds of encryption methods.

In summary, encryption is the most natural way to ensure data confidentiality, which can well improve the security of reader privacy data on the untrustworthy server-side of a digital library. However, up to present, no encryption method can support all kinds of data query operations with acceptable performance overhead. All the encryption methods mentioned above make some trade-offs to different degree in function, security and performance, e.g. (1) for fully homomorphism encryption, it can support all kinds of calculations, but its performance overhead is unacceptable; (2) for semi-homomorphism encryption, its performance overhead is acceptable, but it can only support addition or multiplication; (3) for order-preserving encryption, it can support range query operation, but it may reveal the order information of plain-text data and (4) for searchable encryption, it can support keyword

matching query operation, but although different index constructions have their own characteristics, they cannot support arithmetic operations, and they are only suitable for some specific query requirements. Therefore, existing encryption methods still cannot meet the requirements in terms of accuracy, efficiency and practicality mentioned in [Section 2.1](#). In addition, for most of existing encryption methods, their implementation needs the support of database system, so that the data management service on the server-side of a digital library needs to be changed, which seriously reduces the practicality of this kind of methods.

4. Behavior privacy methods

From the technical point of view, up to present, in the field of library sciences, there is still a lack of systematic and comprehensive research on the privacy protection of digital library readers' behaviors. However, aiming at the protection of user online privacy in an untrustworthy network environment, a number of effective technical methods have been proposed by scholars in the field of information science, where the representative ones include encryption methods, pseudonym methods, obfuscation methods and confusion methods. In this section, we first present a brief introduction on the technical characteristics of each kind of privacy protection methods and then analyze their application limitations in the protection of digital library reader behavior privacy.

4.1 Encryption methods

Here, encryption refers to encrypting the sensitive data related to user service requests, so as to make it invisible to the untrustworthy server-side, and thus achieve the purpose of privacy protection ([Baumeler et al., 2013](#)). Therefore, under the premise of ensuring the availability of information services, encryption can improve the security of user privacy information behind user service requests, consequently, achieving stricter privacy protection. Specifically, encryption methods can be further divided into privacy protection based on privacy information retrieval (PIR protocol) and privacy protection based on cryptographic protocol. The PIR protocol ([Mouratidis et al., 2012](#)) was first used to access user data in an outsourced network environment, which allows users to retrieve their target data from a database on the premise that the server-side cannot know their query requests. It seems that the PIR protocol can also be used to protect user behavior privacy after some modification. However, it is difficult for the PIR protocol to be applied to modern text retrieval because of its high algorithm complexity and its inability of the text retrieval server to carry out targeted text advertising ([Murugesan and Clifton, 2009](#)). In addition, encryption methods based on PIR are also applicable to the location privacy protection in location-based services (LBS). For example, [Ghinita et al. \(2008\)](#) proposed an LBS nearest neighbor query method based on the PIR protocol, and then in [Papadopoulos et al. \(2010\)](#), aiming at k -nearest neighbor queries, which are more general than nearest neighbor queries, the authors designed a PIR protocol requiring the assistance of security hardware. The above two works are targeted only for traditional spatial queries. In addition, there are still some works for other kinds of spatial queries, such as shortest path queries ([Mouratidis et al., 2012](#)). However, the PIR protocol is generally developed based on some cryptographic operations with high complexity, so it can only support limited LBS data access mode ([Wu et al., 2015a, b](#)). The privacy protection methods based on cryptographic protocol are mainly designed for text retrieval, which can be divided into symmetric key encryption based on keyword retrieval ([Ateniese et al., 2012](#)) and public key encryption based on keyword retrieval ([Zhang et al., 2016](#)). After some extension, this kind of methods can help users to retrieve the target text documents that fully meet a given keyword Boolean conditional expression. However, as pointed out in [Pang et al. \(2010a, 2012\)](#), it is difficult for the cryptographic protocol to be applied to modern text retrieval, since modern text retrieval needs to retrieve the most similar text document with a given user

query (i.e. modern text retrieval needs to support similarity text retrieval rather than deterministic retrieval).

In summary, the privacy protection methods based on encryption do not take into consideration the problem of user privacy measurement, so they cannot achieve the complete protection of user privacy. It is more important that this kind of methods not only requires the assistance of additional hardware and high complexity algorithm but also requires the change of information service algorithms on the server-side, consequently, leading to the change of the architecture of an information service platform and greatly reducing the actual usability of this kind of methods in a digital library platform (i.e. they cannot meet the practicality requirement mentioned in Constraint 7).

4.2 Pseudonym methods

The basic idea of pseudonym methods is to replace the user identity in a service request with a temporary pseudonym to break the natural connection between the user and the service request (Gao *et al.*, 2013). A pseudonym method usually adopts a centralized architecture, i.e. publishing, using and revoking pseudonyms through a trusted third-party anonymous server. It is easy for this kind of method to integrate into an existing network service to protect the user behavior privacy, so has been applied in many network information systems (Gao *et al.*, 2013; Sun *et al.*, 2015; Li *et al.*, 2015). However, this kind of method is heavily dependent on the effectiveness of pseudonyms. Therefore, in order to enhance the effectiveness of pseudonyms, a mixed region was proposed in Sun *et al.* (2015) and Li *et al.* (2015a, b), which refers to a specific region where multiple users change their pseudonyms in a centralized way, and the users cannot submit service requests or receive service information, consequently, making it difficult for an attacker to track the users. In Gao *et al.* (2013), a mixed region model based on k -anonymity is further proposed, which requires that the number of users who change the pseudonyms simultaneously in a mixed region is not less than the integer value k , consequently, further improving the effectiveness of user privacy protection. However, users in a mixed region cannot communicate with each other, which inevitably would reduce the information service quality. To solve this, many research studies (Arain *et al.*, 2016; Liu *et al.*, 2012) have proposed their methods for multi-mixed region deployment to achieve a better balance between privacy security and service quality. However, this kind of method has also been questioned. Narayanan and Shmatikov (2008) and Shang *et al.* (2014) analyze the shortcomings of pseudonym methods for user privacy protection by experimental evaluation. The results demonstrate that it is generally difficult to guarantee the quality of user data collected by a pseudonym method, since users may submit useless data without confirming their identity.

As pointed out in Wu *et al.* (2015a, b, 2020b), using pseudonyms alone cannot fully protect user privacy because it does not make any changes to the user service request data (such as query location and query text), which makes attackers still have some probability to determine the user real identity according to the user service requests themselves, i.e. it is difficult for pseudonym methods to resist the attacks from data mining. More importantly, because the user identity is hidden, it is an obstacle for pseudonym methods to an application scenario that requires user identity authentication. In a digital library system, generally, it requires users to login with real name before using various library information services. Therefore, it is difficult to apply a pseudonym method to a digital library system to protect the privacy of readers' behavior preferences (i.e. it cannot meet the practicality requirement mentioned in Constraint 7).

4.3 Obfuscation methods

The basic idea of obfuscation method is to use fake data or general data to obfuscate the behavioral data related to user sensitive preferences (Pang *et al.*, 2010b), making it difficult for

the untrustworthy server-side to accurately know the user sensitive preferences. Aiming at a text retrieval service, Pang *et al.* (2010b) propose to inject pseudo keywords into a user query to protect the user real intention. Then, the authors improve their work in Pang *et al.* (2012), so as to allow users to define their own privacy protection requirements, i.e. allow users to define the query subjects to be protected and the protection level. For personalized advertisement recommendation, a client-based user privacy protection method is proposed in Goetz and Nath (2012), which can select relevant advertisements for users, through taking into account the degree of privacy protection (the level of privacy that users are willing to share with the untrustworthy server-side) and the network data traffic (the number of text advertisements returned by the server-side to phones). In Shou *et al.* (2014), a user preference protection method for personalized web page retrieval services is designed. It first establishes a hierarchical structure of user preferences, where the nodes of high-level are used to store general preference topics, while the low-level nodes are used to store specific preferences. Then, it uses general preferences to replace users' specific preferences, so as to protect the user sensitive preferences. Aiming at personalized web page retrieval services, in Chen *et al.* (2011), a similar preference protection method is proposed, i.e. using general preferences to cover up users' specific preferences so as to protect the sensitive information behind user data. In Murugesan and Clifton (2009), a user privacy protection method for text search is designed. This method constructs several static query groups in advance, and then at run time, each user query is replaced by the most similar static query, while other queries in the same group are used to hide the user query. As pointed out in Wu *et al.* (2015a, b), the main problem of this method is that replacing each user query with a similar static query would reduce the precision performance.

In summary, this kind of method requires no change to the service algorithm on the server-side and no change to the existing service platform architecture, so it has better practicability. However, due to rewriting the data within user service requests, this kind of method would reduce the accuracy of information service to a certain degree, i.e. its privacy protection needs to sacrifice some service quality, which cannot meet the actual application needs of a digital library system.

4.4 Confusion methods

A confusion method is mainly oriented to LBS, which is used to protect users' location privacy, i.e. through generalizing or confusing users' query location information, making it difficult for attackers to identify users' exact query locations (Wu *et al.*, 2020b). Here, generalization refers to replacing users' real locations with some generalized regions (called cloaking regions), which are usually generated by a trusted third-party server in combination with k -anonymity (Damiani, 2014). However, for a traditional cloaking region generation method based on k -anonymity (Chow *et al.*, 2009; Gedik *et al.*, 2008), it is generally difficult to achieve a predetermined privacy protection level in continuous space queries. To this end, some recent research studies attempt to improve this problem. For example, in Lee and Kim (2012), the correlation between users' moving positions is taken into account in the construction of a cloaking region; and the method proposed in Xue *et al.* (2013a, b) can prevent the leakage of the destination location information in a user moving track. In addition, because the privacy needs of users are dynamic and diverse, the impact of personalized privacy needs of users should also be taken into account in the construction of cloaking regions. For example, the method proposed in Agir *et al.* (2014) can adjust its location privacy protection level in a certain range automatically, so as to satisfy users' personalized location privacy requirements in spatial continuous queries; and a user-centered location service framework was proposed in Dewri and Thurimella (2014), which can in advance balance the privacy intensity and practical effect of each user query request. However, a generalized

confusion method is usually dependent on an anonymous third-party server, which reduces the practicability of the method (Wu *et al.*, 2021b). Moreover, confusing users' query location information refers to intentionally adding some errors or noises into user queries in a controllable manner (Dewri and Thurimella, 2016). In order to provide strict privacy protection, recent research studies attempt to apply the well-known differential privacy mechanism to control the amount of noises or errors added in continuous queries, among which the most representative ones are spatial indistinguishability model (Dewri and Thurimella, 2016) and its derivative model (Chatzikokolakis *et al.*, 2015; Zhou *et al.*, 2020). Now, some researchers also attempt to extend confusion methods to the field of information retrieval (Wu *et al.*, 2020c, 2021b).

However, most of existing confusion methods are mainly applicable for the protection of location privacy, which are usually difficult to be directly applied to the reader privacy protection of information services in a digital library (such as information retrieval and recommendation). In addition, because the location information issued to the server-side has been rewritten, the accuracy of LBS would be affected, consequently, reducing the actual usability of this kind of method to a certain extent (i.e. it cannot meet the usability requirement mentioned in Constraints 5–7 and the security requirement mentioned in Constraint 8).

5. Result and discussion

5.1 Data privacy problem

From Section 3, we know that traditional database security technical strategies (such as identity authentication and access control) can effectively prevent unauthorized data access from external users of a system, which seem to be able to be used to solve the problem of reader data privacy in a digital library. However, these strategies are based on an assumption that the server-side of a digital library is highly trusted; thus, they are only targeted for illegal users outside the digital library system and cannot prevent internal managers of the untrustworthy server-side (or attackers who have controlled the server-side) from accessing reader privacy information stored in the library database. Therefore, it is impossible to solve the problem of security of digital library reader data privacy by using these existing database strategies. Data encryption is another immediate way to protect the confidentiality of privacy data in an untrustworthy server-side; thus, it can be used to improve the security of reader data privacy effectively. However, there exist a large number of data query operations defined on reader data in a digital library system. Thus, it is a huge challenge how to realize secure and efficient query operations over the encrypted reader data, i.e. it results in a new problem that needs to be overcome how to ensure the usability of an encryption method. According to the application requirements given in Section 2.1 in terms of security, practicality, accuracy and efficiency, Table 1 presents the comparison result for the two kinds of methods mentioned in Section 3.

Based on the discussion mentioned above, we know that in order to achieve a good balance between security and usability, the following three problems should be mainly taken into consideration in the protection of reader data privacy of a digital library system.

Table 1.
A comparison of effectiveness of all kinds of technical methods

Methods	Accuracy	Efficiency	Security	Practicality I	Practicality II	Practicality III
Security strategies	Good	Good	Not good	Good	Good	Good
Data encryption	Good	Not good	Good	Good	Good	Not good

- (1) Problem 1: We should not only pay attention to the security of library readers' data privacy but also pay attention to the practicality of a privacy protection method. In order to protect the confidentiality of user privacy data, most of existing technical methods (especially for encryption methods) require the cooperation support of a database system deployed on the server-side, so as to implement these technical methods at the database level, which seriously reduces the practical usability of the methods. Therefore, the protection of reader data privacy should be completely transparent to client users and data management services at the server-side, i.e. it is not required to change the usage patterns of client users due to the privacy protection nor to change the background database system, so as to make that the privacy protection approach can be integrated into an existing digital library platform easily.
- (2) Problem 2: We should not only pay attention to the security of library readers' data privacy but also pay attention to the efficiency and effectiveness of a privacy protection method. There are a great number of data query operations defined on reader profile data in a digital library, but existing data encryption methods can only support one specific kind of data query operations, which reduces the availability of the methods. Therefore, in an existing digital library platform, the introduction of a protection method of reader data privacy should not sacrifice the effectiveness of data query operations (i.e. it can support all kinds of query operations and ensure the accuracy of query results) and the efficiency of data query operations.
- (3) Problem 3: To ensure the security of reader privacy data, we should not only pay attention to external attackers of a digital library system but also pay attention to internal managers of the untrustworthy server-side. With the widespread application and popularization of emerging network technologies, such as cloud computing, digital library servers are widely deployed to the cloud, which weakens the control and management of a digital library to reader privacy data, reduces the credibility of the server-side of a digital library and as a result, greatly increases the risk of reader privacy leakage. However, all existing database security technical strategies are generally based on the assumption that the server-sides should be highly trusted, so they cannot be used to ensure the security of reader privacy data under an untrustworthy network environment effectively.

In summary, an effective approach to the protection of reader data privacy should be able to improve the security of reader data privacy on the untrustworthy server-side under the basic precondition of not changing the query efficiency and accuracy nor the usability of an existing digital library system. Although a number of security technical strategies (such as user authentication and access control) are supported in a database system (so they are of good practicability for the protection of reader data privacy), they are developed based on an assumption that the server-side of a digital library is highly trusted, which cannot meet the security constraint of reader data privacy protection. Data encryption is one of the most natural tools to ensure the confidentiality of privacy data, which can well ensure the security of reader data privacy on the untrustworthy server-side of a digital library. However, most of existing encryption methods still cannot meet the accuracy and efficiency constraints, and they usually need to change the data management services on the server-side, so also cannot meet the practicability constraint.

5.2 Behavior privacy problem

From above, we see that existing database security technical strategies can provide some references for the protection of reader data privacy in a digital library. However, the

protection of reader behavior privacy seems to be more complicated, which cannot be solved simply by existing database security technical strategies. From [Section 4](#), we know that although a number of effective user privacy protection methods have been proposed by scholars in the field of computer sciences from the perspective of technology, all of them have their own application limitations in terms of security, accuracy, efficiency and practicality. More importantly, these privacy protection methods are not specifically proposed for a digital library system, consequently, making them still unable to meet the actual needs of a digital library in terms of security, accuracy, efficiency and practicality; thus, it is difficult to be directly applied to the protection of digital library readers' behavior privacy.

According to the above application requirements in terms of security, practicality, accuracy and efficiency, [Table 2](#) presents the comparison result for the four kinds of technical methods mentioned in [Section 4](#). From [Table 2](#), we can see that existing privacy technical methods still cannot meet the application requirements of a digital library system in terms of practicability, accuracy, efficiency and security. Especially for security, most of the existing methods are targeted only for readers' current behaviors, without taking into consideration readers' historical behaviors, and targeted only for a single type of reader behavior privacy, making it difficult to be applied to protect other types of reader behavior privacy. Overall, it is still difficult for existing technical methods to be applied to protect reader behavior privacy of a digital library under an untrustworthy network environment.

Based on the discussion mentioned above, we know that in the protection of reader behavior privacy of a digital library system, the following three problems should be taken into consideration.

- (1) Problem 1: We should not only pay attention to the security of digital library readers' behavior privacy but also pay attention to the accuracy, efficiency and practicality of a privacy protection method. In general, all kinds of knowledge information services (such as retrieval, download, lending and reading) exist as an important part of a large-scale digital library platform, so the platform usability cannot be compromised by behavior privacy protection. However, to protect user privacy, many existing technical methods require changing the existing platform architecture or the service algorithm on the server-side or sacrificing the accuracy and efficiency of information service, as a result, seriously reducing the actual availability of the methods. Therefore, the protection of reader behavior privacy in a digital library should be based on no compromises to the practicality, accuracy and efficiency of existing information services.
- (2) Problem 2: We should not only pay attention to the privacy protection of reader current behaviors (i.e. reader current service requests) but also pay attention to the privacy protection of reader historical behaviors, i.e. the protection of reader behavior privacy should be based on a sequence of reader behaviors (a sequence of service requests submitted by the same reader during a certain period of time). In a digital library system, a sequence of historical behaviors from the same reader exhibits regular characteristic distribution. For example, readers often like to retrieve data

Table 2.

A comparison of effectiveness of all kinds of technical methods

Methods	Accuracy	Efficiency	Practicality	Security I	Security II	Security III
Encryption	Good	Good	Not good	Good	Not good	Not good
Pseudonym	Good	Good	Not good	Good	Not good	Not good
Obfuscation	Not good	Good	Good	Good	Not good	Not good
Confusion	Not good	Good	Good	Good	Not good	Not good

around some fixed topics during a certain period of time. However, most of existing technical methods only focus on reader current behaviors, without taking into account historical behaviors, consequently, seriously reducing the privacy protection effect of the methods. Thus, the protection of reader behavior privacy should be based on a sequence of reader historical behaviors, so as to make it difficult for attackers to infer the reader behavior privacy based on either the current behaviors or the historical behaviors.

- (3) Problem 3: We should not only pay attention to single type of reader behavior privacy but also establish a uniform behavior privacy model to protect all types of reader behavior privacy as a whole. Most of the existing methods are targeted only for single type of behavior privacy (e.g. only for location privacy or retrieval privacy) and lack of comprehensive and systematic. However, information services in a digital library system involve a variety of reader privacy (such as recommendation privacy and retrieval privacy), among which, there are strong semantic correlations. To this end, it is required that we should not take into account the single type of behavior privacy independently. Instead, we should comprehensively consider the semantic correlations among different kinds of behaviors from the same reader and establish a uniform model and framework for reader behavior privacy protection, so as to achieve an effective protection to reader behavior privacy, and thus comprehensively improve the security of reader behavior privacy on the untrustworthy server-side of a digital library system.

In summary, the server-side of a digital library is untrustworthy, which is the main target of attackers and the main source of reader privacy leakage. However, each kind of information service of a digital library cannot be separated from the support of the server-side, and the requests submitted by readers contain a great deal of preference privacy information, consequently, making that in a digital library system, there are a variety of behavior privacy leakage channels. Therefore, an ideal approach to the protection of digital library readers' behavior privacy should meet the following application goals, i.e. under the premises of not changing the existing platform architecture and service algorithms (i.e. practicality) nor the information service accuracy and efficiency, it should be difficult for attackers of the server-side to analyze and infer reader behavior privacy from the service request sequences issued by readers, so as to comprehensively improve the security of reader behavior privacy.

6. Conclusions

The problem of reader privacy has become one of the important obstacles to the further development and application of a digital library platform. This paper reviews a large number of research achievements related to user privacy protection, analyzes and evaluates their application limitations in the protection of digital library reader privacy and establishes the constraints that an effective approach to the protection of reader data privacy and behavior privacy should satisfy, so as to provide references and directions for the follow-up study of this problem. At present, the protection of reader privacy has two main directions, i.e. based on laws and regulations and based on technical methods. However, according to the analysis of this paper, we have conclusions as follows.

- (1) Although laws and regulations can effectively alleviate the problem of reader privacy, existing laws and regulations are still lacking systematic and comprehensive research, and the endless emergence of privacy leakage events shows that making laws and regulations cannot fundamentally solve the reader privacy problem.

- (2) Although a database provides many effective security technical strategies, they are all built based on the assumption that the server-side is highly trusted, which cannot meet the actual privacy protection needs of digital libraries in an untrustworthy network environment.
- (3) Although aiming at the problem of user privacy in an untrustworthy network environment, in the field of computer sciences, there have been many effective research achievements proposed from the perspective of technical methods, they have their own limitations and there are still some problems that need to be solved. Moreover, these technical methods are not specifically proposed for a digital library system, so they still cannot meet the actual needs of reader privacy protection of a digital library in terms of security, accuracy, efficiency and practicality, i.e. it is difficult to directly apply them for the reader privacy protection in a digital library system.

Finally, based on the analysis to the shortcomings of existing technical methods, in combination with the practical application requirements of a digital library system, this paper has a conclusion as follows. An ideal approach to the protection of digital library reader privacy should be able to comprehensively improve the security of all kinds of reader privacy on the untrustworthy server-side, without changing the existing platform architecture, the information service algorithm on the server-side and the accuracy and efficiency of information service.

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